Autism and Communication

AAC AND AUTISM: A GUIDE FOR EDUCATORS

Introduction

All people have fundamental communication rights to engage in communication interactions (Brady, 2016). Approximately 30% of autistic children do not use verbal speech as their primary form of communication by the time they reach school age (Tager-Flusberg & Kasari, 2013). Students with complex communication needs may need augmentative or alternative ways to communicate (AAC). Finding the most effective and efficient AAC system and supporting AAC users and partners empowers students to spontaneously communicate whatever they desire with any partner in any environment. This guide will help educators answer frequent questions related to AAC and autism.

Identity-first language is used throughout this tipsheet. For more information access the TRIAD identify language web page (triad.vumc.org/identity-language).

Learn more about the National Joint Committee Communication Bill of Rights (asha.org/siteassets/njc/njc-communication-bill-rights.pdf).

What is AAC?

Augmentative and alternative communication (AAC) refers to using another form of communication to supplement or replace spoken words. AAC is under the umbrella of assistive technology. Any communication other than an established language [verbal (e.g., English) or signed (e.g., American Sign Language or ASL)] is considered augmentative and alternative communication. AAC is multimodal, encouraging students to incorporate every mode possible to share information. AAC includes gestures, facial expressions, eye gaze, writing, and voice output devices.

> An AAC **system** encompasses all methods of communication whereas an AAC **device** refers to the tool (e.g., voice output device, pictures, communication book) assisting with communication.

AAC is a tool in the same way a pencil is a tool. It does not automatically grant the individual the ability to communicate, but rather can help **facilitate** their communication in an effective and efficient manner.

For more information on assistive technology, access this Tennessee Department of Education resource (tn.gov/content/dam/tn/education/special-education/Assistive_Technology_2021.pdf).

How will AAC affect speech development?

There is no evidence that the use of AAC slows down speech development. Conversely, there is growing evidence that using AAC facilitates the development of effective speech, social and academic skills. For more information access this Communication Community web page (communicationcommunity.com/does-aac-prevent-speech-development/).

Who may benefit from AAC?

Any person whose daily communication needs are not met, including young children who require support with their speech and/or language development, may benefit from AAC. AAC can be used effectively with autistic students who are nonspeaking from early childhood through high school (Franzone, 2008). AAC helps people of all ages and there are no prerequisite skills.

How might AAC benefit my autistic student?

AAC may give a student a way to express wants and needs, engage socially, and develop communication skills. Providing a student with a means for clear communication may also help reduce frustration with guessing and misinterpretation. In addition, AAC may decrease challenging behaviors from students when their current forms of communication are misunderstood.

How long will my student use AAC?

For young AAC users, it is impossible to predict which students will develop speech independent of AAC and which students will benefit from some type of AAC system indefinitely. The decision to continue utilizing AAC is based on the student's communicative, academic, and social strengths and needs. For students who continue using AAC, AAC systems are modified to match their changing communication needs.

Are there different types of AAC?

Yes, there are many distinct types of AAC systems. Some are “no-tech” and do not require anything beyond the user’s body. Others are rapid access and require something external to the user that may be non-electronic (i.e., paper-based) or a simple electronic device (i.e., “mid-tech”). “High-tech” types of AAC are electronic devices similar to computers.

AAC Options

What are the different types of AAC systems?

Remember many individuals will use a combination of communication supports throughout their day and all communication is honored. There is no one “best” AAC intervention for all autistic children. The key is matching your student’s individual strengths and needs with the most appropriate type(s) of AAC. Some students respond best to a combination of different types of AAC.

Types of AAC systems

Name	Description
No-tech	No-tech methods of communication may include head nods, gestures, and eye gaze.
Rapid Access/ Paper-Based systems	Rapid Access/Paper-Based systems do not need electricity or batteries and can be used quickly in a variety of settings. These communication systems may contain any of the following: pictures, symbols, letters, words, or phrases placed upon paper, single page board or a printed book.
Picture Exchange Communication System (PECS)	You may have heard of the Picture Exchange Communication System (PECS), which is a specific communication protocol where the individual exchanges pictures to communicate a message. PECS often focuses primarily on fringe vocabulary and does not build into a robust language system.
Mid-tech systems	Mid-tech systems have a power source (usually a battery) and produce a message when a button is pressed. These communication systems may include a single message button like a Big Mac or contain a few prerecorded messages such as a Go Talk or Quick Talker.
High-tech systems	High-tech systems require a power source (usually charged via electrical outlet) that allows for storage and retrieval of prestored messages that can be used alone or in combination to produce robust communication. High-tech systems may include dedicated devices or iPads with a communication app.

Questions about high-tech AAC devices:

"There are so many words on this system! Can words be removed?"
Before editing the student's system by limiting the words, decide if this will limit communication in settings or with partners. It is important to presume competence in students so that we do not limit a student's potential. Frequently limiting the words by masking or changing the grid size does not help a student learn a new method of communication quicker, but only holds them back from communicating what they want in a variety of contexts. Consider how many different requests you have made today. Would you feel successful in those interactions if your communication was limited to only a few words? More words are often better than fewer words.

"I want a page for a specific activity, should a page be added?"
Most of the high-tech communication software programs have thousands of words preprogrammed. Instead of creating a page to talk about snacks, or one specific game, try to teach the individual how to use the search feature or use similar words to communicate. These strategies will help them throughout their communication journey, as everyone uses search features and descriptive communication.

Visual Supports are any visuals (pictures, photographs, objects, words) used to support the student. Examples include visual schedules, timers, task breakdowns, feelings charts, or first/then boards. Often visual supports alone are not considered AAC if they are not accompanied by other behaviors to effectively communicate (e.g., pointing to the visual, handing the visual to another person, etc.). However, visual supports are often used in combination with AAC systems. They can support internal language, such as talking to yourself through your daily schedule, and therefore are great supports for those with executive functioning difficulties.

Communication Evidence-Based Practices

When do students need access to AAC devices?

Students should have access to AAC devices in all environments. Students using vocal communication can communicate with others at all times, through speech. **A student with an AAC device has the same right to always have access to communication**.

Who teaches students to use AAC devices?

Everyone! Educators, paraprofessionals, parents, and peers can work collaboratively to teach the student. AAC users need to see other people using AAC.

While AAC users may receive direct support from therapy providers, related services a few times per week will not result in the AAC user being successful in the classroom. AAC users need classroom teachers and paraprofessionals to model language using the AAC system throughout the day. This type of communication strategy is referred to as aided language stimulation.

For more information on aided language stimulation, explore the professional learning modules on the Project Core website (see resources).

What strategies promote communication for AAC users?

Many of the strategies are used with all students throughout the school day. To improve the effectiveness and efficiency of communication, AAC users need communication partners to be more intentional about using these best practices.

Strategies to Promote Communication for AAC Users

Strategy	Guidance
Wait time	Wait time is one of the most important strategies. Allowing twice the time provides AAC users to process what is being said to them and provide a response.
Model	Model language on the AAC user's system. Build communication skills using the following strategies and choosing key words to model on the AAC system. Use a Variety of Words Teach language using a variety of words. Think about commonly used words in a student's everyday speech. The words "my," "want" and "go" are much more commonly used than "swing," "cookie" or "toilet." Core words and fringe words are both important. Attribute Meaning Assign meaning to facial expressions, vocalizations, body movements or gestures. If the student moves their body away when they do not like something, say "I see you moving away, you do ***not like*** that." Self-Talk Say aloud what you are doing. This models your thinking process and internal language. This is a great strategy to use when looking for a word to model on the AAC device. "Hmmm, I'm looking for the word "huge." I wonder where it could be. Maybe under ***descriptions***? Ooo, I see cups of three different sizes, that makes me think it may be size words and "huge" is a word that tells us the size. Let's see if "huge" is in that folder!". Learning from others' mistakes is powerful learning, so it's ok to be wrong! Parallel Talk Say aloud what the AAC user is doing. Be specific and give them the language to talk about what they are engaged with. For example: "You're ***making*** a train track. That train is going ***over*** the bridge. Now it's ***next*** to the tree. The train is going really ***fast***."

Strategy	Guidance
Comment	Join your communication user in the activity by commenting on their actions or what is happening around them, with no expectation of a response. Strive to make more comments and ask fewer questions. Comments may sound like, "The ***ball*** is ***red***. It's really bouncy. Whoa! ***It goes up*** so high!"
Recast/ Expand	Take the AAC user's communication attempt and provide an enhanced and/or contrasting version of the utterance by changing one or more sentence components. If the student selects the word "go," the expansion and recast may sound like, "You really ***want*** to **go**. I'm sorry, we can ***not go*** until after lunch."
Prompt	Provide the least amount of prompting and wait before providing additional visual or verbal prompts. Start with the above modeling strategies without expectation of a response along with adequate wait time. If further cues are needed, start with an indirect visual cue, such as moving the AAC system closer or gesturing towards the device. Prompt Hierarchy Before using hand-over-hand prompting, use less intensive types of prompts. Refrain from physical prompts as much as possible as they can lead to learned helplessness and to a lack of bodily autonomy. Ensure student consent prior to any type of physical assistance. Hand-under-hand allows students to remain in control of their own body. Consider hand-under-hand for those students who need that type of support. For more information on the prompt hierarchy, access this resource (https://bit.ly/44AsdBT).

Make learning language fun and engaging! You are doing amazing things in your classroom. Use what you already are doing in your classroom and incorporate AAC into those activities.

Resources

TRIAD (triad.vumc.org), the autism institute at Vanderbilt Kennedy Center, and the Assistive Technology Project (tn-tan.tnedu.gov/support-services/assistive-technology), are proud members of the Tennessee Technical Assistance Network (TN-TAN) through the Tennessee Department of Education. TN-TAN provides school districts, administrators, educators, and families access to high-quality training, resources, and supports designed to improve outcomes for students with disabilities, ages 3-22. For general inquiries, email TN-TAN (TN-TAN@utk.edu).

Project Core is an implementation grant directed by the Center for Literacy and Disability Studies. The Project Core implementation model is designed to empower teachers and classroom professionals to provide students with access to flexible Universal Core vocabulary and evidence-based communication instruction.

- Quick Start Guide (project-core.com/quick-start-guide)
- Professional Development Modules (project-core.com/professional-development-modules)

References

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Tager-Flusberg, H., & Kasari, C. (2013). Minimally verbal school-aged children with autism spectrum disorder: The neglected end of the spectrum. *Autism Research, 6*(6), 468–478. https://doi.org/10.1002/aur.1329

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